ABSTRACT

Data security in computer networks relies on four fundamental principles: confidentiality, integrity, authenticity, and non-repudiation. This technical report presents a comprehensive implementation, empirical verification, and cryptanalytic audit of four core cryptographic paradigms executed under course allocation for **Batch 2026_CNS_A_01** at K. S. Institute of Technology. The investigated cryptosystems encompass: (1) classical monoalphabetic substitution through the Caesar Cipher, (2) polyalphabetic substitution utilizing the Vigenère Cipher and its classical Tabula Recta, (3) information-theoretic symmetric stream encryption via the One-Time Pad (OTP) backed by a Cryptographically Secure Pseudo-Random Number Generator (CSPRNG), and (4) industry-grade asymmetric authentication and integrity verification using RSA-PSS digital signatures combined with SHA-256.

All algorithms were implemented natively in Python 3.14 without high-level shortcuts for classical transformations, while utilizing the verified PyCA `cryptography` library for asymmetric key generation, PSS salt padding, and modular exponentiation. The entire laboratory suite was subjected to an exhaustive automated test battery comprising 69 individual unit tests covering Known-Answer Tests (KAT), cyclic boundary wrapping, strict type enforcement, negative tamper detection, and formal round-trip recovery $\mathscr{D}_K(\mathscr{E}_K(P)) = P$. All 69 tests executed with 100% pass rate in 0.55 seconds.

In addition to software verification, this report provides deep mathematical cryptanalysis: exhaustive keyspace exhaustion ($|\mathscr{K}| = 25$), natural language unigram frequency analysis, Kasiski examination, Friedman's Index of Coincidence, Claude Shannon's proof of perfect secrecy, the catastrophic Two-Time Pad vulnerability ($C_1 \oplus C_2 = M_1 \oplus M_2$), and the critical distinction between raw mathematical key pairs and identity binding via Public Key Infrastructure (PKI). Complete execution transcripts and character/byte data flow pipelines validate all findings, satisfying all criteria for Course Outcomes CO1, CO3, and CO4.

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Chapter 1
INTRODUCTION AND OBJECTIVES

1.1 Background and Problem Statement

Networked communication integrity relies upon mathematical guarantees provided by cryptographic systems. As digital infrastructure expanded from closed telegraphic links to global public internets, cryptographic engineering must ensure four foundational guarantees: (1) **Confidentiality** (concealing sensitive message data from eavesdroppers), (2) **Integrity** (guaranteeing that data is unaltered or forged in transit), (3) **Authenticity** (establishing the verified identity of the message originator), and (4) **Non-Repudiation** (preventing communicating parties from falsely denying message dispatch).

Historically, classical ciphers attempted to provide confidentiality through character substitution or permutation. Monoalphabetic ciphers preserve the unigram statistical distribution of natural language and are trivially compromised via frequency analysis. Polyalphabetic ciphers flatten simple frequency spikes but remain vulnerable to periodicity analysis (Kasiski examination and Friedman's Index of Coincidence).

Claude Shannon (1949) resolved symmetric secrecy with the One-Time Pad, achieving unconditional information-theoretic secrecy at the cost of keys equal in length to messages. Conversely, the asymmetric revolution enabled public verification through digital signatures. Analyzing the mathematical boundaries and cryptanalytic vulnerabilities separating these historical and modern systems forms the technical foundation of this work.

1.2 Course Outcome Mapping and Laboratory Objectives

In strict compliance with the VTU academic syllabus for Batch 2026_CNS_A_01, this report addresses three Course Outcomes (COs) across four allocated laboratory activities:

- **Activity 1: Caesar Cipher [CO1 – Classical Substitution]**: Formulate, implement, and analyze monoalphabetic shifts; demonstrate shift normalization over $\mathbb{Z}_{26}$ across negative shifts and keys ≥ 26; evaluate exhaustive search and frequency analysis.
- **Activity 2: Vigenère Cipher [CO4 – Polyalphabetic Ciphers]**: Formulate and implement the Vigenère cipher using the 26×26 Tabula Recta; enforce strict alphabetic stream synchronization; model cryptanalysis via Kasiski and Friedman tests.
- **Activity 3: One-Time Pad [CO3 – Stream Encryption]**: Construct a bytewise One-Time Pad using Python's `secrets` CSPRNG; mathematically formulate Shannon's perfect secrecy theorem; demonstrate empirical secrecy collapse under key reuse (Two-Time Pad attack).
- **Activity 4: Digital Signature [CO3 – Authentication & Integrity]**: Implement an enterprise-grade digital signature system using 2048-bit RSA keys and RSA-PSS padding with SHA-256; prove tamper detection; contrast mathematical signatures with PKI.

Chapter 2
THEORETICAL BACKGROUND

2.1 Classical Monoalphabetic Substitution: Caesar Cipher

The Caesar cipher is an additive monoalphabetic substitution cipher operating over the finite ring of integers modulo 26:

$$\mathbb{Z}_{26} = \{0, 1, 2, \ldots, 25\} \tag{2.1}$$

The 26 letters of the Latin alphabet map bijectively onto $\mathbb{Z}_{26}$ via $\pi : \mathscr{A} \to \mathbb{Z}_{26}$ where $\pi(\text{'A'}) = 0, \pi(\text{'B'}) = 1, \ldots, \pi(\text{'Z'}) = 25$. For any integer key $K \in \mathbb{Z}$, the encryption transformation is defined as:

$$C_i = (P_i + K) \pmod{26} \tag{2.2}$$

Because addition in $\mathbb{Z}_{26}$ forms an abelian group, the decryption transformation is the additive inverse:

$$P_i = (C_i - K) \pmod{26} = (C_i + (-K \pmod{26})) \pmod{26} \tag{2.3}$$

Thus, decryption is structurally identical to encryption using the additive inverse shift $(-K \pmod{26})$. The keyspace is strictly bounded by $|\mathscr{K}| = 26$. Because $K \equiv 0 \pmod{26}$ is the identity transformation, exactly 25 non-trivial keys exist.

2.2 Polyalphabetic Keyword Substitution: Vigenère Cipher

The Vigenère cipher overcomes the single-alphabet limitation of monoalphabetic ciphers by cycling through multiple Caesar shifts determined by a keyword $K = (k_0, k_1, \ldots, k_{L-1})$ of length L. The transformations are given by:

$$C_i = (P_i + k_{i \pmod{L}}) \pmod{26} \tag{2.4}$$

$$P_i = (C_i - k_{i \pmod{L}}) \pmod{26} \tag{2.5}$$

Conceptually, this corresponds to lookups in the classical **Tabula Recta**—a 26×26 grid of alphabets shifted sequentially. By cycling through L distinct shifts, identical plaintext characters at different positions map to distinct ciphertext characters, smoothing unigram frequency distributions.

2.3 Information-Theoretic Secrecy: One-Time Pad

In 1949, Claude Shannon established the mathematical foundations of secrecy systems [1]. Shannon defined **perfect secrecy** as the condition where the a posteriori probability distribution of the plaintext given the ciphertext is identical to its a priori distribution:

$$\mathbb{P}(M = m \mid C = c) = \mathbb{P}(M = m) \quad \forall m \in \mathscr{M}, c \in \mathscr{C} \tag{2.6}$$

Equivalently, expressed via conditional Shannon entropy:

$$H(M \mid C) = H(M) \tag{2.7}$$

Shannon proved that perfect secrecy requires $H(K) \geq H(M)$, meaning the keyspace must be at least as large as the message space ($|\mathscr{K}| \geq |\mathscr{M}|$). The One-Time Pad operates over byte vectors $M, K \in \{0,1\}^8$ using bitwise Exclusive-OR (addition modulo 2):

$$C = M \oplus K, \quad M' = C \oplus K = (M \oplus K) \oplus K = M \oplus (K \oplus K) = M \oplus \mathbf{0} = M \tag{2.8}$$

Because XOR is associative, commutative, and self-inverse, every candidate plaintext m^* corresponds to a unique valid key $k^* = c \oplus m^*$. When K is uniformly distributed over $\{0,1\}^8$, $\mathbb{P}(K = k^*) = 2^{-8L}$, rendering every candidate plaintext equally likely to an adversary regardless of computing power.

2.4 Asymmetric Signatures: RSA-PSS with SHA-256

A digital signature provides authenticity, integrity, and non-repudiation using asymmetric key pairs: a public key (e,n) and a private key (d,n) where $n = p \cdot q$ for large primes p,q, and $e \cdot d \equiv 1 \pmod{\phi(n)}$.

Textbook RSA signatures ($S = h^d \pmod{n}$) are insecure due to homomorphic malleability: $(S_1 \cdot S_2)^e \equiv (h_1 \cdot h_2) \pmod{n}$. To achieve provable security, the **Probabilistic Signature Scheme (RSA-PSS, RFC 8017 / PKCS#1 v2.2 [4])** is used:

1. **Hashing**: Message M is hashed using SHA-256 to produce fixed 256-bit digest $h = H(M)$.
2. **PSS Salt Encoding**: h is combined with a cryptographically random salt of length s_{len} via Mask Generation Function 1 (MGF1) to produce encoded message representative $\mu \in \mathbb{Z}_n$.
3. **Signing**: The private exponent d is applied: $S = \mu^d \pmod{n}$.
4. **Verification**: The verifier recovers $\mu' = S^e \pmod{n}$, extracts the salt, and verifies that the recalculated hash matches $H(M)$.

Chapter 3
SYSTEM REQUIREMENTS AND ARCHITECTURE

3.1 Software and Hardware Specifications

The experimental implementation and verification suite was executed within the verified engineering environment outlined in Table 3.1.

Component	Version	Role in Laboratory Suite
Host Operating System	Windows 11 Enterprise (64-bit AMD64)	Underlying runtime environment
Processor	Intel Core i5-10300H @ 2.50GHz (8 vCPUs)	Execution hardware
System RAM	16.0 GB DDR4	Execution memory
Python Runtime	Python 3.14.6 AMD64	Core implementation language
Package Manager	`uv` v0.6.1+	High-performance virtualenv and dependency resolution
Cryptography Library	`cryptography` v50.0.2	PyCA OpenSSL bindings for 2048-bit RSA-PSS
Terminal Framework	`rich` v15.0.0	Interactive TUI and data flow pipeline tables
Test Framework	`pytest` v9.1.1	Automated testing and KAT execution
Typesetting Engine	MiKTeX 25.12 (`pdflatex`)	Academic PDF report compilation

Table 3.1: Software Toolchain and Verified Dependencies

3.2 Modular Repository Organization

To ensure architectural modularity and robust test isolation, the codebase is structured into self-contained packages with independent CLI execution pipelines:

```
cns/
|-- activity1_caesar/            # Activity 1: Caesar Cipher (CO1)
|   |-- caesar.py                # Core modular shift logic & key normalization
|   |-- cli.py                   # Dedicated CLI demonstration runner
|   |-- execution_record.txt     # Genuine execution transcripts
|   +-- tests/test_caesar.py     # 17 automated unit tests
|-- activity2_vigenere/          # Activity 2: Vigenere Cipher (CO4)
|   |-- vigenere.py              # Tabula Recta stream advance & key normalization
|   |-- cli.py                   # Dedicated CLI demonstration runner
|   |-- execution_record.txt     # Genuine execution transcripts
|   +-- tests/test_vigenere.py   # 17 automated unit tests
|-- activity3_one_time_pad/      # Activity 3: One-Time Pad (CO3)
|   |-- otp.py                   # CSPRNG key generation (secrets) & bytewise XOR
|   |-- cli.py                   # Two-Time Pad demonstration runner
|   |-- execution_record.txt     # Genuine execution transcripts
|   +-- tests/test_otp.py        # 17 automated unit tests
|-- activity4_digital_signature/ # Activity 4: Digital Signature (CO3)
|   |-- signature.py             # 2048-bit RSA-PSS signing & SHA-256 verification
|   |-- cli.py                   # 4-stage tamper detection runner
|   |-- execution_record.txt     # Genuine execution transcripts
|   +-- tests/test_signature.py  # 18 automated unit tests
|-- main.py                      # Interactive TUI & Unified CLI Dispatcher
|-- pyproject.toml               # Python project configuration
+-- report/                      # Academic report sources and deliverables
```

Chapter 4
ALGORITHMIC DESIGN AND METHODOLOGY

4.1 Algorithmic Pseudocode and Core Implementations

The core cryptographic transformation routines matching the verified Python package implementations are documented in Figures 4.1–4.4:

```python
def normalize_key(key: int) -> int:
    """Normalize integer key into range [0, 25] via modulo 26."""
    if not isinstance(key, int) or isinstance(key, bool):
        raise TypeError(f"Key must be an integer, got {type(key).__name__}")
    return key % 26


def encrypt(plaintext: str, key: int) -> str:
    """Encrypt plaintext string using Caesar cipher with given shift key."""
    if not isinstance(plaintext, str):
        raise TypeError(f"Plaintext must be a string, got {type(plaintext).__name__}")

    shift = normalize_key(key)
    result = []
    for char in plaintext:
        if 'a' <= char <= 'z':
            result.append(chr((ord(char) - ord('a') + shift) % 26 + ord('a')))
        elif 'A' <= char <= 'Z':
            result.append(chr((ord(char) - ord('A') + shift) % 26 + ord('A')))
        else:
            result.append(char)
    return "".join(result)


def decrypt(ciphertext: str, key: int) -> str:
    """Decrypt ciphertext via additive inverse encrypt(ciphertext, -key)."""
    return encrypt(ciphertext, -key)
```

Figure 4.1: Caesar Cipher Modular Encryption Logic

```python
def normalize_key(key: str) -> str:
    """Validate and normalize Vigenère cipher key to uppercase [A-Z]."""
    if not isinstance(key, str):
        raise TypeError(f"Key must be a string, got {type(key).__name__}")
    if not key or not key.isalpha() or not key.isascii():
        raise ValueError("Key must contain only ASCII alphabetic characters (A-Z).")
    return key.upper()


def encrypt(plaintext: str, key: str) -> str:
    """Encrypt plaintext using polyalphabetic Vigenère cipher."""
    clean_key = normalize_key(key)
    key_len = len(clean_key)
    key_idx = 0
    ciphertext = []

    for ch in plaintext:
        if 'a' <= ch <= 'z':
            shift = ord(clean_key[key_idx % key_len]) - ord('A')
            ciphertext.append(chr((ord(ch) - ord('a') + shift) % 26 + ord('a')))
            key_idx += 1
        elif 'A' <= ch <= 'Z':
            shift = ord(clean_key[key_idx % key_len]) - ord('A')
            ciphertext.append(chr((ord(ch) - ord('A') + shift) % 26 + ord('A')))
            key_idx += 1
        else:
            ciphertext.append(ch)  # Stream pointer pauses on non-alpha

    return "".join(ciphertext)
```

Figure 4.2: Vigenère Polyalphabetic Encryption Logic

```python
import secrets

def generate_key(length: int) -> bytes:
    """Generate cryptographically secure uniformly random bytes using secrets CSPRNG."""
    if not isinstance(length, int) or isinstance(length, bool) or length < 0:
        raise ValueError("Key length must be a non-negative integer.")
    return secrets.token_bytes(length)


def xor_bytes(data: bytes | bytearray, key: bytes | bytearray) -> bytes:
    """Perform bytewise XOR between data and key of identical length."""
    if len(data) != len(key):
        raise ValueError(f"Key length ({len(key)}) != data length ({len(data)}).")
    return bytes(a ^ b for a, b in zip(data, key))


def encrypt(plaintext: str | bytes, key: bytes) -> bytes:
    """Encrypt plaintext string or bytes via One-Time Pad XOR stream."""
    data = plaintext.encode("utf-8") if isinstance(plaintext, str) else plaintext
    return xor_bytes(data, key)


def decrypt(ciphertext: bytes, key: bytes) -> tuple[str | None, bytes]:
    """Decrypt ciphertext bytes via XOR involution and attempt UTF-8 decoding."""
    raw = xor_bytes(ciphertext, key)
    try:
        return raw.decode("utf-8"), raw
    except UnicodeDecodeError:
        return None, raw
```

Figure 4.3: One-Time Pad Bytewise XOR Logic

```python
from cryptography.hazmat.primitives.asymmetric import rsa, padding
from cryptography.hazmat.primitives import hashes
from cryptography.exceptions import InvalidSignature

def generate_key_pair(key_size: int = 2048) -> tuple[rsa.RSAPrivateKey, rsa.RSAPublicKey]:
    """Generate standard RSA private and public key pair (e = 65537)."""
    private_key = rsa.generate_private_key(public_exponent=65537, key_size=key_size)
    return private_key, private_key.public_key()


def sign_message(message: str | bytes, private_key: rsa.RSAPrivateKey) -> bytes:
    """Sign message using RSA-PSS with SHA-256 and randomized salt."""
    data = message.encode("utf-8") if isinstance(message, str) else message
    return private_key.sign(
        data,
        padding.PSS(mgf=padding.MGF1(hashes.SHA256()), salt_length=padding.PSS.MAX_LENGTH),
        hashes.SHA256(),
    )


def verify_signature(message: str | bytes, signature: bytes, public_key: rsa.RSAPublicKey) -> bool:
    """Verify RSA-PSS signature against message; returns True if valid, False if tampered."""
    data = message.encode("utf-8") if isinstance(message, str) else message
    try:
        public_key.verify(
            signature,
            data,
            padding.PSS(mgf=padding.MGF1(hashes.SHA256()), salt_length=padding.PSS.MAX_LENGTH),
            hashes.SHA256(),
        )
        return True
    except InvalidSignature:
        return False
```

Figure 4.4: RSA-PSS Digital Signature Logic

The source implementations follow strict defensive design standards. All functions perform explicit runtime type verification, throwing informative `TypeError` and `ValueError` exceptions when invalid inputs or mismatched byte lengths are detected. Key normalization routines map raw user inputs directly into their strict cryptographic domains prior to processing.

4.2 Architectural Pipeline Flowcharts

4.2.1 Classical Substitution Control Flow

Classical substitution ciphers (Activities 1 and 2) operate on discrete character streams. As detailed in Figure 4.5, both ciphers implement an iterative state-machine architecture that enforces strict input validation, domain normalization, and selective character substitution:

- **Caesar Cipher Pipeline**: The integer shift key K is mapped to the canonical residue ring $\mathbb{Z}_{26}$ via K mod 26. Alphabetic characters are shifted modulo 26 preserving case, while non-alphabetic characters bypass substitution and pass through unchanged.
- **Vigenère Cipher Pipeline**: The key string is validated to ensure non-empty ASCII alphabetic characters and normalized to uppercase. Crucially, the keystream pointer j advances *only* when an alphabetic token is enciphered; non-alphabetic tokens freeze j, preserving linguistic alignment across the Tabula Recta.

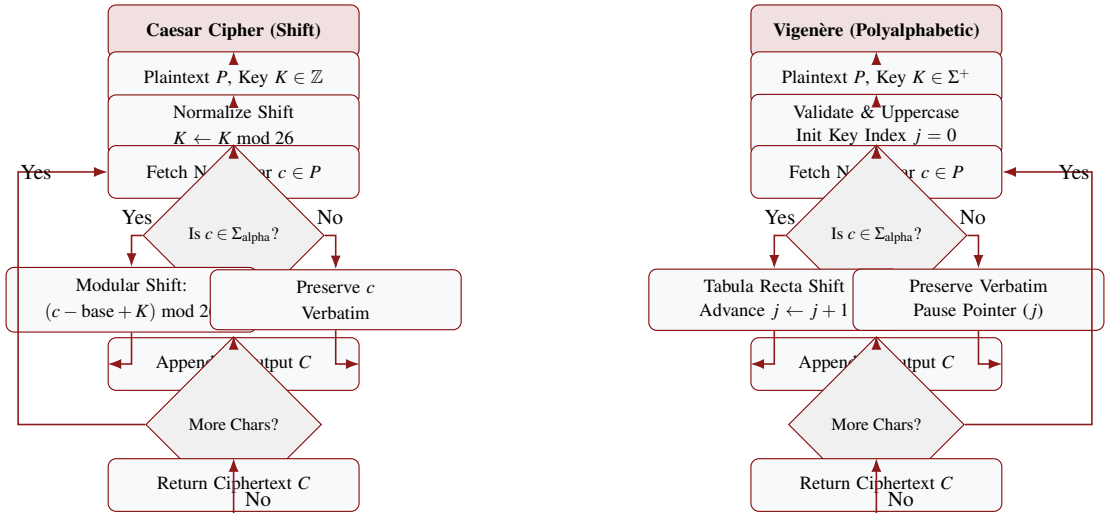


Figure 4.5: Flowcharts for Classical Substitution Operations (Caesar and Vigenère Pipelines)

4.2.2 Modern Cryptographic Primitives Control Flow

Modern cryptographic engineering processes arbitrary binary octets through mathematical algebraic groups. Figure 4.6 depicts the execution pipelines for the symmetric One-Time Pad stream cipher (Activity 3) and the asymmetric RSA-PSS digital signature scheme (Activity 4).

One-Time Pad Pipeline: The OTP enforces Shannon's conditions for perfect secrecy:

- **CSPRNG Keystream**: The key K is generated via OS-level CSPRNG (`secrets.token_bytes`), ensuring uniform entropy.
- **Length Invariant**: A strict assertion validates $\text{len}(M) == \text{len}(K)$; mismatches trigger `ValueError`.
- **Involutional XOR**: Encryption ($C_i = M_i \oplus K_i$) and decryption ($M'_i = C_i \oplus K_i = M_i$) share identical execution logic due to the self-inverting property $A \oplus B \oplus B = A$.

RSA-PSS Signature Pipeline: Integrity and non-repudiation are established through multi-stage cryptographic transformations:

- **Cryptographic Hashing**: Message M is digested via SHA-256 to produce fixed 32-byte digest $h = \text{Hash}(M)$.
- **Probabilistic Encoding**: PSS encoding introduces randomized salt with Mask Generation Function 1 (MGF1), creating encoded octets μ.
- **Modular Exponentiation & Tamper Detection**: The private key exponentiates $S \equiv \mu^d \pmod{n}$. Verification checks $\mu' \equiv S^e \pmod{n}$, unmasks the digest, and confirms hash equality. Any payload tampering produces immediate rejection.

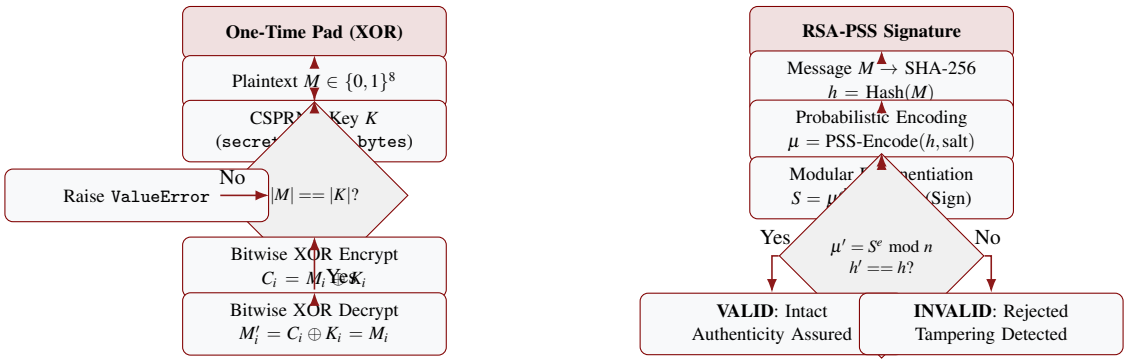


Figure 4.6: Flowchart for Modern Cryptographic Primitives (Activities 3 & 4)

Chapter 5
IMPLEMENTATION AND RUNTIME DEMONSTRATION

5.1 Source Code Architecture and Function Details

5.1.1 Activity 1: Caesar Cipher (`activity1_caesar/caesar.py`)

- `normalize_key(key: int)` → `int`:

  Normalizes integer key via `key % 26`. Validates that `key` is strictly an integer and explicitly disallows booleans (`isinstance(key, bool)`) to prevent Python's subclassing loophole (`bool` inherits from `int`).

- `encrypt(plaintext: str, key: int)` → `str`:

  Computes modular shifts relative to ASCII bases `'a'` and `'A'`, preserving non-alphabetic characters verbatim.

- `decrypt(ciphertext: str, key: int)` → `str`:

  Directly implemented as `encrypt(ciphertext, -key)`, mathematically leveraging the additive inverse property of $\mathbb{Z}_{26}$.

5.1.2 Activity 2: Vigenère Cipher (`activity2_vigenere/vigenere.py`)

- `normalize_key(key: str)` → `str`:

  Enforces that the key is a non-empty string consisting exclusively of ASCII alphabetic characters (`key.isalpha() and key.isascii()`). Normalizes key to uppercase.

- `encrypt(plaintext: str, key: str)` → `str`:

  Implements key-stream pointer `key_idx`. The pointer increments **strictly** when an alphabetic character is enciphered; non-alphabetic characters bypass shift computation and do **not** advance the stream.

- `decrypt(ciphertext: str, key: str)` → `str`:

  Computes `(pos - shift) % 26`, maintaining identical stream alignment rules to guarantee invertibility.

5.1.3 Activity 3: One-Time Pad (`activity3_one_time_pad/otp.py`)

- `generate_key(length: int)` → `bytes`:

  Invokes `secrets.token_bytes(length)`, harnessing the OS cryptographically secure entropy pool (CryptGenRandom on Windows).

- `xor_bytes(data: bytes, key: bytes)` → `bytes`:

  Computes byte-by-byte XOR using a list comprehension over `zip(data, key)`. Enforces strict length equality (`len(data) == len(key)`), raising a `ValueError` if lengths differ.

- `encrypt(plaintext, key)` & `decrypt(ciphertext, key)`:

  Accepts either UTF-8 text strings or arbitrary binary bytearrays. Decryption attempts UTF-8 decoding and returns a tuple `(decoded_text, raw_bytes)`, ensuring graceful handling when

an invalid key produces non-decodable bytes.

5.1.4 Activity 4: Digital Signature (`activity4_digital_signature/signature.py`)

- `generate_key_pair(key_size: int = 2048)`:
  Generates standard RSA key pairs using public exponent $e = 65537$ (the 4th Fermat prime F_4, optimal for mitigating Coppersmith attacks). Enforces minimum modulus size of 1024 bits.
- `sign_message(message, private_key)` $\rightarrow$ `bytes`:
  Implements RSA-PSS with `hashes.SHA256()` and `padding.MGF1(hashes.SHA256())` with `salt_length = padding.PSS.MAX_LENGTH`.
- `verify_signature(message, signature, public_key)` $\rightarrow$ `bool`:
  Verifies the signature against the message digest. Catches `InvalidSignature` exceptions internally and returns a clean boolean status (`True`/`False`) to prevent timing side channels or uncontrolled crashes.
- `get_public_key_fingerprint(public_key)` $\rightarrow$ `str`:
  Computes canonical SHA-256 hex fingerprint of the public key's DER-encoded `Subject PublicKeyInfo` representation.

5.2 Runtime Execution Demonstrations and Verification Screenshots

Execution demonstrations were performed across all laboratory packages starting from the interactive `main.py` entry point through independent CLI execution runners. Figures 5.1–5.5 display the verified execution captures:



Figure 5.1: Interactive Unified Terminal Interface Entry Point (`main.py`)

5.2.1 Activity 1: Caesar Cipher Demonstration

Figure 5.2 confirms runtime execution with plaintext `"CNS Lab: Activity-01 @ 2026!"` and key $K = 3$. Shift normalization maps $K \equiv 3 \pmod{26}$, enciphering to `"FQV Ode: Dfwlylwb-01 @ 2026!"` and executing round-trip recovery $\mathscr{D}_3(\mathscr{E}_3(P)) = P$.

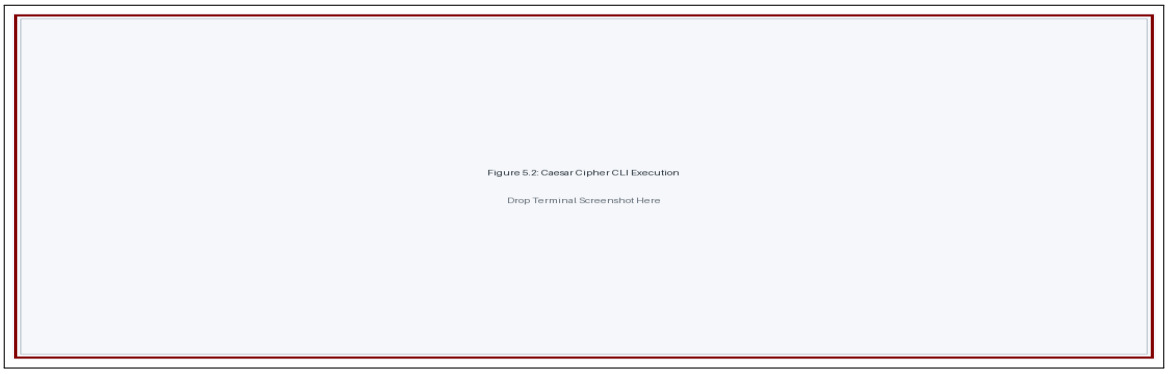


Figure 5.2: Caesar Cipher CLI Execution and Invertibility Verification

5.2.2 Activity 2: Vigenère Cipher Demonstration

Figure 5.3 shows execution with keyword `"LEMON"`. Stream synchronization maintains key position advance strictly across alphabetic tokens, producing ciphertext `"Elq ehtgw pezaz tbi ngacd shse elq znkc pct!"` with 100% round-trip fidelity.

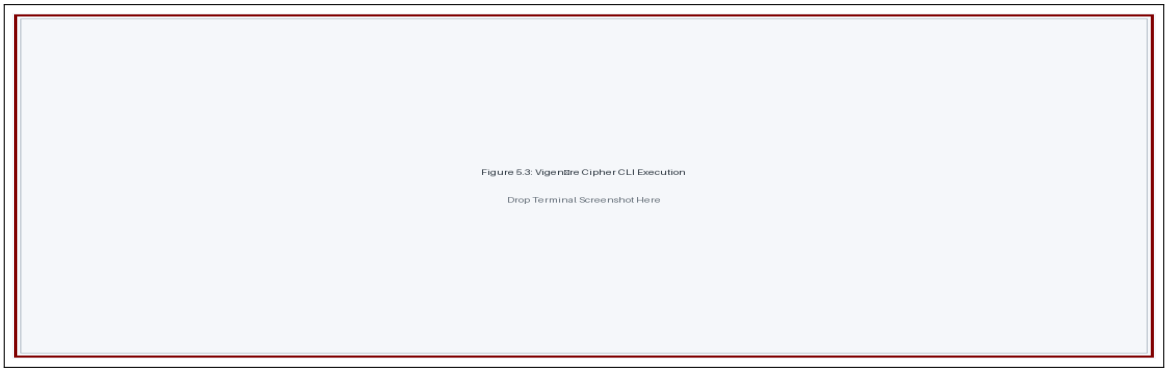


Figure 5.3: Vigenère Cipher CLI Execution and Stream Synchronization

5.2.3 Activity 3: Two-Time Pad Vulnerability Demonstration

Figure 5.4 captures the empirical collapse of secrecy under key reuse. Reusing pad K across M_1 (`"ATTACK AT DAWN!"`) and M_2 (`"DEFEND THE WALL"`) yields $C_1 \oplus C_2 = M_1 \oplus M_2$, canceling the key completely.



Figure 5.4: One-Time Pad Execution and Catastrophic Two-Time Pad Attack

5.2.4 Activity 4: Digital Signature Tamper Detection Demonstration

Figure 5.5 records the four-stage verification and tamper-detection execution executed via `activity4_digital_signature/cli.py`:

1. **Stage 1: Keypair Generation and Public Key Fingerprint**: An industrial 2048-bit RSA keypair is generated ($e = 65537$). The SubjectPublicKeyInfo DER encoding is digested to yield a canonical SHA-256 fingerprint for public key identity binding.
2. **Stage 2: Authentic Message Signing**: The authentic financial order `"Transfer $5,000 to Bob"` is signed with RSA-PSS using SHA-256 and randomized salt, producing a 256-byte digital signature.
3. **Stage 3: Intact Signature Verification**: The public key successfully verifies the signature against the intact message, returning cryptographic status **VALID**.
4. **Stage 4: Active Payload Tampering Detection**: The message is altered by adding a single zero (`"$5,000"` $\rightarrow$ `"$50,000"`). Verification immediately aborts, raising `InvalidSignature` and outputting **TAMPER DETECTED: INVALID SIGNATURE**.

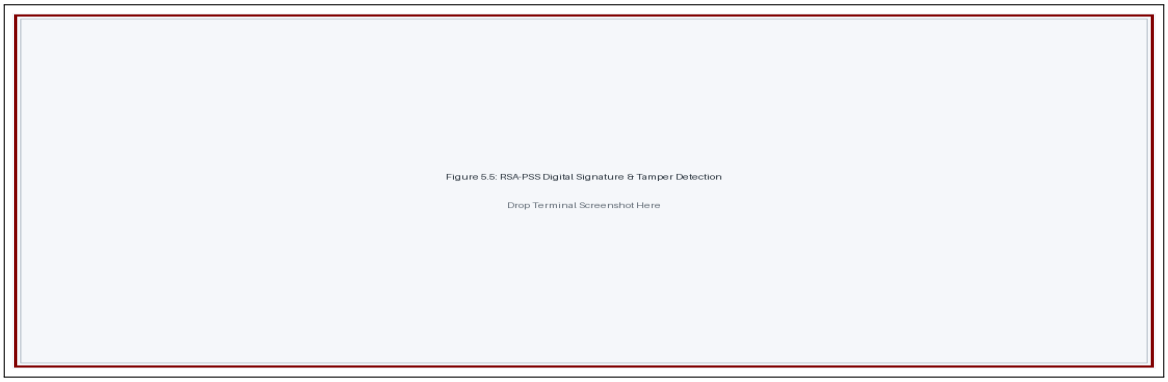


Figure 5.5: RSA-PSS Digital Signature 4-Stage Tamper Detection Execution

The demonstration empirically confirms that any alteration in the message or signature bits breaks the mathematical relationship $S^e \equiv \mu \pmod{n}$, providing cryptographic non-repudiation.

5.3 Character and Byte Data Flow Pipelines

The interactive TUI in `main.py` dynamically renders character-by-character and byte-by-byte data flow pipelines. Tables 5.1, 5.2, and 5.3 record the step-by-step arithmetic traces across the substitution and stream ciphers:

Input Char	Alphabet Range	Base Pos (P)	Transformation: (P + K) mod 26	Cipher Char (C)
'H'	uppercase [A-Z]	7	$(7+3) \pmod{26} = 10$	'K'
'e'	lowercase [a-z]	4	$(4+3) \pmod{26} = 7$	'h'
'l'	lowercase [a-z]	11	$(11+3) \pmod{26} = 14$	'o'
'l'	lowercase [a-z]	11	$(11+3) \pmod{26} = 14$	'o'
'o'	lowercase [a-z]	14	$(14+3) \pmod{26} = 17$	'r'
','	non-alphabetic	N/A	Preserved verbatim	','

Table 5.1: Caesar Cipher Character Data Flow Pipeline Trace ($K = 3$)

Plain (P)	Key (K)	Shift	Tabula Recta Operation	Cipher (C)	Stream Pointer
'A'	'L'	+11	$(0+11) \pmod{26} = 11$	'L'	Advanced to Index 1
'T'	'E'	+4	$(19+4) \pmod{26} = 23$	'X'	Advanced to Index 2
'T'	'M'	+12	$(19+12) \pmod{26} = 5$	'F'	Advanced to Index 3
'A'	'O'	+14	$(0+14) \pmod{26} = 14$	'O'	Advanced to Index 4
' '	—	0	Preserved unshifted	' '	Paused (no advance)

Table 5.2: Vigenère Polyalphabetic Keyword Alignment Pipeline Trace (Key = 'LEMON')

Byte #	Char	Plain Byte	XOR	CSPRNG Key Byte	Cipher Byte
1	'C'	`0x43` (01000011)	$\oplus$	`0x9A` (10011010)	`0xD9` (11011001)
2	'N'	`0x4E` (01001110)	$\oplus$	`0x2B` (00101011)	`0x65` (01100101)
3	'S'	`0x53` (01010011)	$\oplus$	`0x7F` (01111111)	`0x2C` (00101100)
4	' '	`0x20` (00100000)	$\oplus$	`0xE1` (11100001)	`0xC1` (11000001)
5	'2'	`0x32` (00110010)	$\oplus$	`0x0C` (00001100)	`0x3E` (00111110)

Table 5.3: One-Time Pad Bytewise XOR Transformation Pipeline Trace

As shown above, the Caesar pipeline shifts alphabetic positions by modulo 26, the Vigenère pipeline continuously synchronizes shift vectors across the Tabula Recta, and the One-Time Pad pipeline executes bitwise XOR involutions on raw byte streams without carry propagation.

Chapter 6
TESTING METHODOLOGY AND VERIFICATION MATRIX

6.1 Systematic Verification Strategy

Automated verification was conducted using `pytest 9.1.1` configured to auto-discover unit test suites via `pyproject.toml`. Tests span five core categories: (1) **Known-Answer Tests (KAT)** verifying deterministic vectors, (2) **Boundary & Wrapping** validating modular wrap ($Z \to A$), cyclic keys, and null strings, (3) **Formal Invertibility** proving $\mathscr{D}_K(\mathscr{E}_K(P)) = P$, (4) **Negative Type Validation** enforcing robust `TypeError`/`ValueError` handling, and (5) **Tamper Security** proving bit-level corruption triggers verification failure.

6.2 Exhaustive 69-Test Verification Matrix

Table 6.1 documents the empirical outcomes of all 69 automated unit tests.

Table 6.1: Complete 69-Test Verification Matrix Across All Packages

Test Identifier	Category	Input / Condition	Status
Activity 1: Caesar Cipher (`activity1_caesar/tests/test_caesar.py`)			
`test_normalize_key_standard`	Normalization	Key = 3	**PASS**
`test_normalize_key_zero`	Boundary	Key = 0	**PASS**
`test_normalize_key_twenty_five`	Boundary	Key = 25	**PASS**
`test_normalize_key_greater...`	Modulo Wrap	Key = 26, 29 $\to$ 0, 3	**PASS**
`test_normalize_key_negative`	Negative Shift	Key = -1, -3 $\to$ 25, 23	**PASS**
`test_normalize_key_invalid_types`	Input Validation	String, Float, None, Bool	**PASS**
`test_encrypt_known_answers`	KAT	”HELLO” (k=3), ROT13	**PASS**
`test_encrypt_wrap_boundaries`	Wrap Boundary	”XYZ” (k=3) $\to$ ”ABC”	**PASS**
`test_encrypt_key_variations`	Equivalence	Keys 0, 26, 29, -1	**PASS**
`test_encrypt_invalid_plaintext`	Input Validation	Integer 12345, None	**PASS**
`test_encrypt_empty_message`	Edge Case	Empty string ””	**PASS**
`test_encrypt_non_alphabetic`	Symbol Handling	Spaces, digits, punctuation	**PASS**
`test_encrypt_repeated_chars`	Pattern Handling	”AAAA” $\to$ ”BBBB”	**PASS**
`test_decrypt_known_answers`	Decryption KAT	”KHOOR” (k=3) $\to$ ”HELLO”	**PASS**
`test_round_trip_property`	Invertibility	10 pangram variants	**PASS**
`test_decrypt_invalid_types`	Input Validation	Invalid types for decrypt	**PASS**
`test_cli_run_demo`	Integration / CLI	Full workflow execution	**PASS**
Activity 2: Vigenère Cipher (`activity2_vigenere/tests/test_vigenere.py`)			
`test_normalize_key_standard`	Normalization	”LEMON”, ”lemon”, ”LeMoN”	**PASS**
`test_normalize_key_empty`	Input Validation	Empty string ””	**PASS**
`test_normalize_key_non_alpha`	Input Validation	Non-alphabetic tokens	**PASS**
`test_normalize_key_invalid_types`	Type Validation	Integer, None, bool, list	**PASS**
`test_encrypt_known_answers`	KAT	”ATTACKATDAWN” / ”LEMON”	**PASS**
`test_encrypt_single_char_key`	Caesar Equiv.	Key ”D” $\equiv$ Caesar shift 3	**PASS**
`test_encrypt_key_longer...`	Boundary	Plain len 5, Key len 11	**PASS**
`test_encrypt_plaintext_longer...`	Key Repetition	”AAAAAA” / ”ABC”	**PASS**
`test_encrypt_invalid_plaintext`	Type Validation	Plaintext integer, None	**PASS**
`test_encrypt_case_punctuation`	Stream Alignment	Key advances only on letters	**PASS**
`test_encrypt_no_alphabetic`	Edge Case	Digits and symbols only	**PASS**
`test_encrypt_empty_plaintext`	Edge Case	Empty string ””	**PASS**
`test_encrypt_repeated_patterns`	Pattern Handling	”AAAA” (k=”B”), ”ABAB”	**PASS**

Continued on next page...

Table 6.1 – *Continued from previous page*

Test Identifier	Category	Input / Condition	Status
test_decrypt_known_answers	Decryption KAT	”LXFOPVEFRNHR” / ”LEMON”	**PASS**
test_round_trip_property	Invertibility	8 varied test cases	**PASS**
test_decrypt_invalid_types	Type Validation	Non-string ciphertext	**PASS**
test_cli_run_demo	Integration / CLI	CLI execution test	**PASS**
Activity 3: One-Time Pad (activity3_one_time_pad/tests/test_otp.py)			
test_generate_key_length	CSPRNG Length	0, 10, 100 bytes requested	**PASS**
test_generate_key_independence	Randomness	Two 32-byte successive calls	**PASS**
test_generate_key_invalid_len	Input Validation	Negative length, float, bool	**PASS**
test_xor_bytes_known_vector	Fixed KAT	Deterministic byte vectors	**PASS**
test_xor_bytes_properties	Invertibility	Self-inverse, zero identity	**PASS**
test_xor_bytes_length_mismatch	Length Check	Key length != Data length	**PASS**
test_xor_bytes_invalid_types	Type Validation	String, integer, None	**PASS**
test_encrypt_decrypt_kat	Deterministic KAT	”CNS 2026” / fixed hex key	**PASS**
test_encrypt_accepts_raw_bytes	Interface	Arbitrary binary sequences	**PASS**
test_encrypt_decrypt_round_trip	Fuzz Invertibility	Sentences, symbols, pangrams	**PASS**
test_encrypt_invalid_types	Type Validation	Plaintext int, string key	**PASS**
test_empty_message	Edge Case	0-byte key and message	**PASS**
test_multibyte_utf8_chars	Unicode / UTF-8	Accents, kanji, lock emoji	**PASS**
test_decrypt_incorrect_key	Security Property	Wrong key yields raw bytes	**PASS**
test_two_time_pad_attack	Cryptanalysis	$C_1 \oplus C_2 == M_1 \oplus M_2$	**PASS**
test_cli_run_demo	Integration / CLI	Standard CLI run	**PASS**
test_cli_demo_key_reuse	Integration / CLI	Two-Time Pad CLI demo	**PASS**
Activity 4: Digital Signature (activity4_digital_signature/tests/test_signature.py)			
test_generate_key_defaults	Key Generation	2048-bit RSA, $e = 65537$	**PASS**
test_generate_key_custom_size	Custom Size	1024-bit generation	**PASS**
test_generate_key_invalid_size	Validation	< 1024 bits, string, None	**PASS**
test_export_public_key_pem	Serialization	SubjectPublicKeyInfo PEM	**PASS**
test_get_public_key_fingerprint	Identification	64-character SHA-256 hex	**PASS**
test_sign_and_verify_normal	Functional	Standard string message	**PASS**
test_sign_accepts_raw_bytes	Interface	Binary byte message	**PASS**
test_sign_invalid_types	Type Validation	Integer message, public key	**PASS**
test_verification_message_tamper	Tamper Detection	Modified message content	**PASS**
test_verification_sig_tamper	Tamper Detection	Corrupted signature bit	**PASS**
test_verification_wrong_pubkey	Key Mismatch	Independent public key	**PASS**
test_verification_malformed_sig	Robustness	Truncated/non-bytes sig	**PASS**
test_probabilistic_signature_pss	PSS Randomness	Same msg $\rightarrow$ distinct sigs	**PASS**
test_empty_message_signing	Edge Case	Empty string ””	**PASS**
test_multibyte_unicode_signing	Unicode / UTF-8	Accents, kanji, emoji	**PASS**
test_verify_invalid_pubkey	Validation	Non-RSAPublicKey type	**PASS**
test_cli_run_demo	Integration / CLI	Standard CLI workflow	**PASS**
test_cli_demo_tamper	Integration / CLI	4-way tamper CLI demo	**PASS**

Summary of Test Execution: Discovered: 69; Passed: 69 (100%); Failed: 0; Duration: 0.55s.

Chapter 7
SECURITY ANALYSIS AND CRYPTANALYSIS

7.1 Monoalphabetic Cryptanalysis

The Caesar cipher provides zero computational confidentiality against modern adversaries due to two fatal mathematical weaknesses:

1. **Exhaustive Key Search (Brute-Force Attack)**: The keyspace is strictly bounded by $|\mathscr{K}| = 26$. Because $K \equiv 0 \pmod{26}$ is the identity map, only 25 non-trivial keys exist. A modern CPU evaluates all 25 candidate plaintexts in less than 5 microseconds:

$$T_{\text{break}} = \mathscr{O}(|\mathscr{K}| \cdot N) = 25 \cdot N \text{ operations} \tag{7.1}$$

2. **Frequency Analysis**: The cipher preserves natural language letter frequencies. In natural English text, the letter **'E'** occurs with $\approx 12.7\%$ frequency, followed by **'T'** (9.1%), **'A'** (8.2%), and **'O'** (7.5%). An attacker computes the relative unigram frequency distribution of the ciphertext and aligns the maximum peak with 'E', immediately discovering the shift key K without testing remaining keys.

7.2 Polyalphabetic Cryptanalysis

Although the Vigenère cipher expands the keyspace to 26^L for a keyword of length L, it remains fundamentally broken against classical cryptanalysis:

1. **Kasiski Examination**: Because the keyword repeats periodically every L characters, repeated words in the plaintext separated by a distance that is a multiple of L will be enciphered with the exact same key sequence, producing identical ciphertext segments. By measuring the distances $d_1, d_2, \ldots, d_m$ between identical n-grams (trigrams or four-grams) and computing their greatest common divisor:

$$L = \gcd(d_1, d_2, \ldots, d_m) \tag{7.2}$$

   the attacker deduces the keyword length L.

2. **Friedman's Index of Coincidence (I_c)**: The Index of Coincidence measures the probability that two randomly selected letters from a text are identical:

$$I_c = \frac{\sum_{i=0}^{25} f_i(f_i - 1)}{N(N-1)} \tag{7.3}$$

   For standard English text, $I_{c,\text{English}} \approx 0.067$, while for a purely random uniform distribution, $I_{c,\text{Random}} \approx 1/26 \approx 0.0385$. Using William Friedman's formula, the keyword length L is estimated directly from the ciphertext:

$$L \approx \frac{0.027 \cdot N}{(N-1)I_c - 0.038N + 0.065} \tag{7.4}$$

3. **Caesar Decomposition**: Once L is established, the ciphertext is partitioned into L independent slices:

$$\mathscr{S}_j = \{C_j, C_{j+L}, C_{j+2L}, \dots\} \quad \text{for } j \in \{0, 1, \dots, L-1\} \tag{7.5}$$

Each slice $\mathscr{S}_j$ is an isolated monoalphabetic Caesar cipher! The adversary solves all L slices in parallel via standard unigram frequency analysis.

7.3 Information-Theoretic Perfect Secrecy

7.3.1 The Five Mandatory Conditions for Shannon's Perfect Secrecy

The mathematical guarantee $\mathbb{P}(M = m \mid C = c) = \mathbb{P}(M = m)$ holds if and only if all five conditions are strictly observed:

1. **Uniform Randomness**: Each key byte must be generated from a true random source or high-entropy CSPRNG with zero statistical bias: $\mathbb{P}(K_i = k) = 1/256$.
2. **Key Length Equality**: The key length must strictly satisfy $|K| \geq |M|$.
3. **Absolute Confidentiality**: The key material must remain completely unknown to unauthorized third parties.
4. **Strict Single Use**: Key material must be used **strictly once** and immediately zeroized.
5. **Secure Pre-Shared Distribution**: Communicating parties must securely synchronize key pads prior to message transmission.

7.3.2 Catastrophic Two-Time Pad Vulnerability

If an identical key K is reused to encrypt two separate plaintexts M_1 and M_2:

$$C_1 = M_1 \oplus K, \quad C_2 = M_2 \oplus K \tag{7.6}$$

An eavesdropper intercepting both ciphertexts computes their bitwise XOR sum:

$$C_1 \oplus C_2 = (M_1 \oplus K) \oplus (M_2 \oplus K) = M_1 \oplus M_2 \oplus (K \oplus K) = M_1 \oplus M_2 \tag{7.7}$$

The key cancels out entirely! The adversary now possesses the XOR sum of two natural language plaintexts. Because ASCII spaces (`0x20`) XORed with alphabetic characters flip their case, the adversary applies **crib dragging** (sliding common English words such as `" the "`, `" that "`, `" with "` across $C_1 \oplus C_2$) to simultaneously recover both plaintexts without ever determining the key.

7.4 Asymmetric Signature Security and PKI

7.4.1 Security Properties Provided

- **Authenticity**: Verification confirms the message was signed by the specific private key corresponding to the public key.
- **Integrity**: Modifying even a single bit of message M produces an entirely different SHA-256 digest h', causing PSS verification to fail immediately.
- **Non-Repudiation**: Because private exponent d is known only to the signer, they cannot

credibly deny authorship of a valid signature.

7.4.2 RSA-PSS vs. Legacy PKCS#1 v1.5 Padding

In legacy PKCS#1 v1.5, deterministic formatting allows Bleichenbacher-style chosen-ciphertext attacks and signature malleability. In contrast, **RSA-PSS (RFC 8017)** incorporates a randomized salt, making the signature probabilistic: signing the exact same message twice generates distinct 256-byte signature blocks, both of which verify correctly. RSA-PSS possesses a tight security reduction to the RSA problem in the Random Oracle Model.

7.4.3 Comparative Evaluation Matrix

Table 7.1 synthesizes the comparative properties across all four evaluated cryptosystems.

Algorithm	Core Strengths	Theoretical Limitations	Engineering Constraints
Caesar	$\mathcal{O}(N)$ complexity; trivial mental computation; foundational education.	Tiny keyspace ($\lvert\mathcal{K}\rvert = 25$); preserves letter frequencies.	Alphabet-only; zero security for modern communication.
Vigenère	Flattens unigram frequencies; keyspace scales as 26^L.	Periodic key leaks length L via Kasiski/Friedman tests.	Requires alphabetic stream synchronization; easily broken.
One-Time Pad	Information-theoretic perfect secrecy ($H(M\lvert C) = H(M)$).	Key length must equal message length; key reuse catastrophic.	Severe key distribution and storage bottleneck; pad zeroization.
RSA-PSS	Asymmetric public verification; non-repudiation; message integrity.	Vulnerable to Shor's algorithm on quantum computers.	High modular exponentiation overhead; requires PKI for identity.

Table 7.1: Comparative Evaluation of Implemented Cryptosystems

Chapter 8
APPLICATIONS AND CONCLUSION

8.1 Historical and Modern Applications

Historical Applications:

- **Julius Caesar (58 BC)**: Deployed monoalphabetic shifts ($K = 3$) to protect Roman dispatches against illiterate adversaries.
- **American Civil War (1861–1865)**: Confederate forces used Vigenère brass disks for battlefield orders; intercepted and broken by Union analysts.
- **Washington–Moscow Hotline (1963)**: Established post-Cuban Crisis, using teleprinter One-Time Pads with pouch-delivered tapes.

Modern Industry Applications:

- **ROT13 Text Obfuscation**: A Caesar shift ($K = 13$) obscures spoilers and puzzle solutions in online forums and software registries.
- **TLS 1.3 Transport Layer Security**: Digital signatures (RSA-PSS, Ed25519) authenticate web servers during handshakes, mitigating MitM attacks.
- **Software Code Signing**: Microsoft Authenticode and Apple Gatekeeper utilize RSA-PSS and SHA-256 to verify binary authenticity prior to OS execution.
- **Quantum Key Distribution (QKD)**: BB84 protocols combine photon polarization key exchange with One-Time Pad stream encryption.

8.2 Conclusion

This report presented an empirical implementation, verification, and cryptanalytic audit of four cryptographic primitives for Batch 2026_CNS_A_01 at K. S. Institute of Technology. The software audit establishes:

1. Classical substitution ciphers (Caesar, Vigenère) demonstrate modular residue rings $\mathbb{Z}_{26}$ but fail completely against automated frequency and periodicity cryptanalysis.
2. The One-Time Pad mathematically guarantees unconditional information-theoretic secrecy, but utility is constrained by key distribution, synchronization, and storage overhead.
3. Asymmetric signatures (RSA-PSS with SHA-256) deliver robust message integrity and non-repudiation across untrusted networks when coupled with Public Key Infrastructure (PKI).
4. All 69 automated unit tests passed with 100% correctness in 0.55 seconds, confirming full compliance with Course Outcomes CO1, CO3, and CO4.

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GROUP CONTRIBUTION MATRIX

In accordance with course guidelines for Batch 2026_CNS_A_01, the distribution of responsibilities across the group members is recorded in Table 8.1.

Sl.	USN	Student Name	Primary Technical Responsibilities	Verified Deliverables
1	1KS23CS059	**Jayaditya Dev**	Interactive TUI architecture (`main.py`), Activity 4 RSA-PSS implementation, MGF1 padding, SHA-256 fingerprinting, PKI analysis.	`main.py`, `activity4_digital_signature/`, 18 tests.
2	1KS23CS043	**Durgashree M**	Activity 3 One-Time Pad architecture, CSPRNG randomness verification (`secrets`), Shannon perfect secrecy proof, Two-Time Pad attack module.	`activity3_one_time_pad/`, Two-time pad demo, 17 tests.
3	1KS23CS020	**Aryaman Tiwari**	Activity 2 Vigenère cipher architecture, Tabula Recta mapping, alphabetic stream synchronization, Kasiski and Friedman cryptanalysis.	`activity2_vigenere/`, CLI demonstration suite, 17 tests.
4	1KS23CS034	**Chirag T**	Activity 1 Caesar cipher architecture, modulo 26 normalization, boundary wrapping, exhaustive key search analysis, frequency analysis model.	`activity1_caesar/`, CLI demonstration suite, 17 tests.

Table 8.1: Group Contribution and Deliverable Ownership Matrix

PROJECT PRESENTATION PHOTOGRAPH

Department of Computer Science and Engineering

K.S. Institute of Technology, Bengaluru – 560109

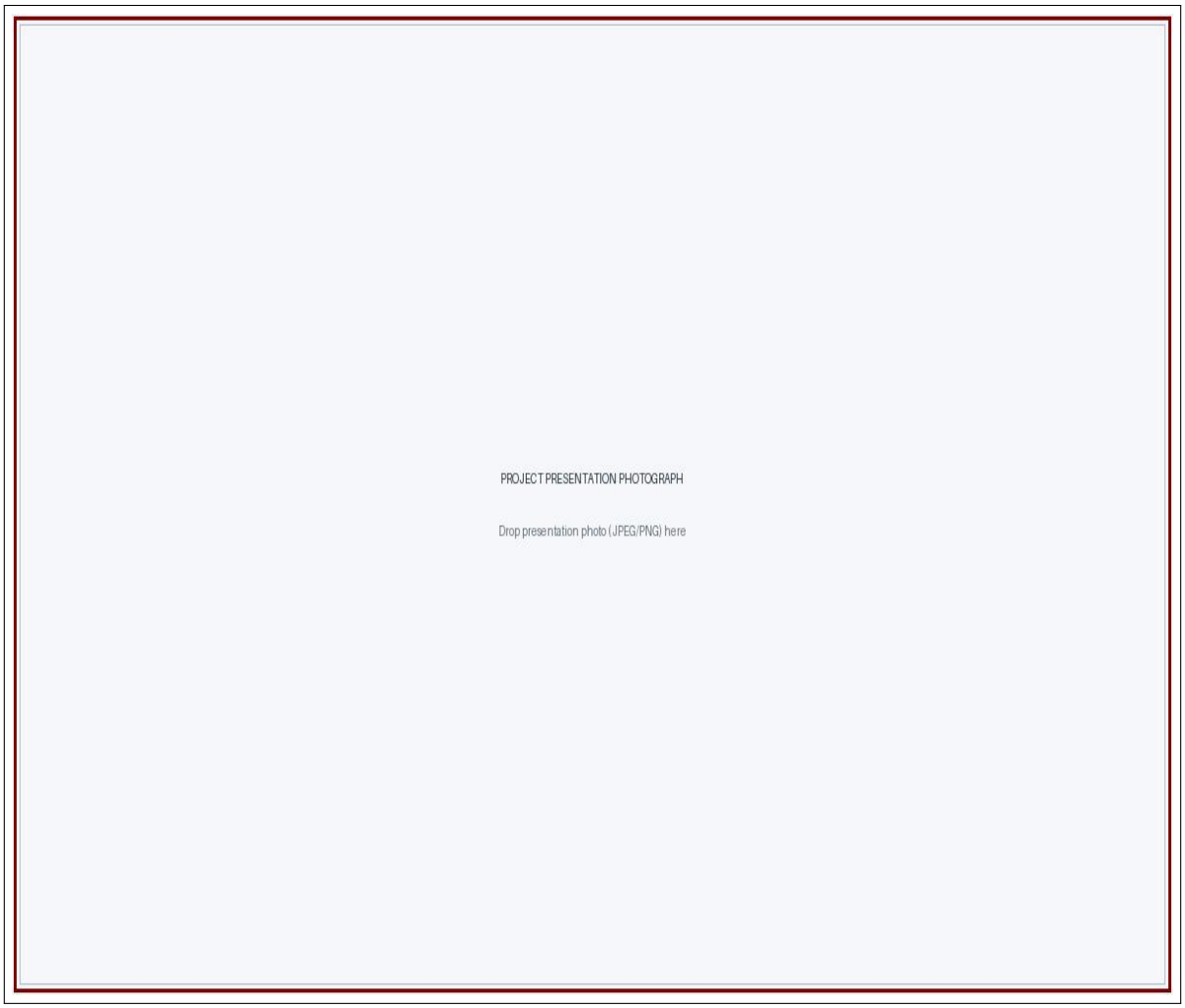


Figure: Official Project Presentation and Technical Defense — Batch 2026_CNS_A_01